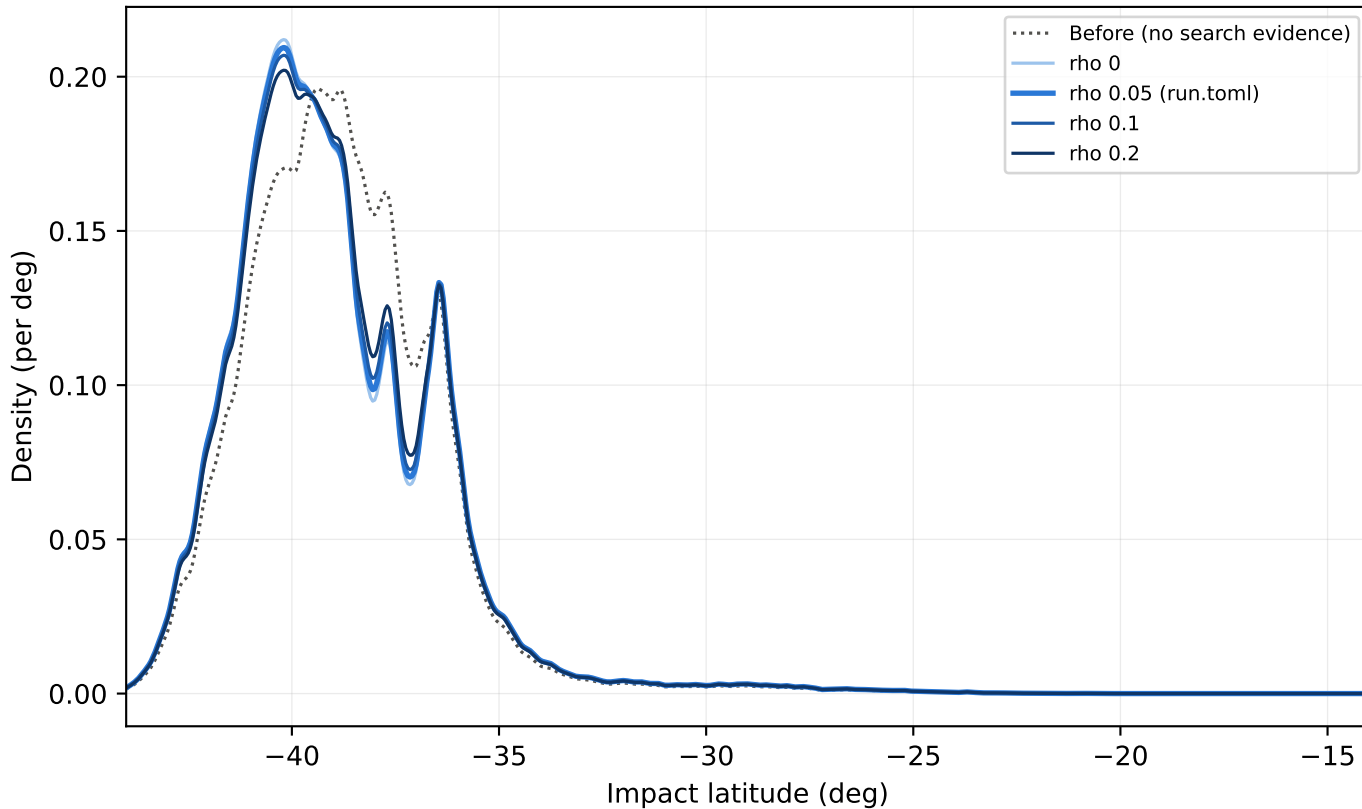
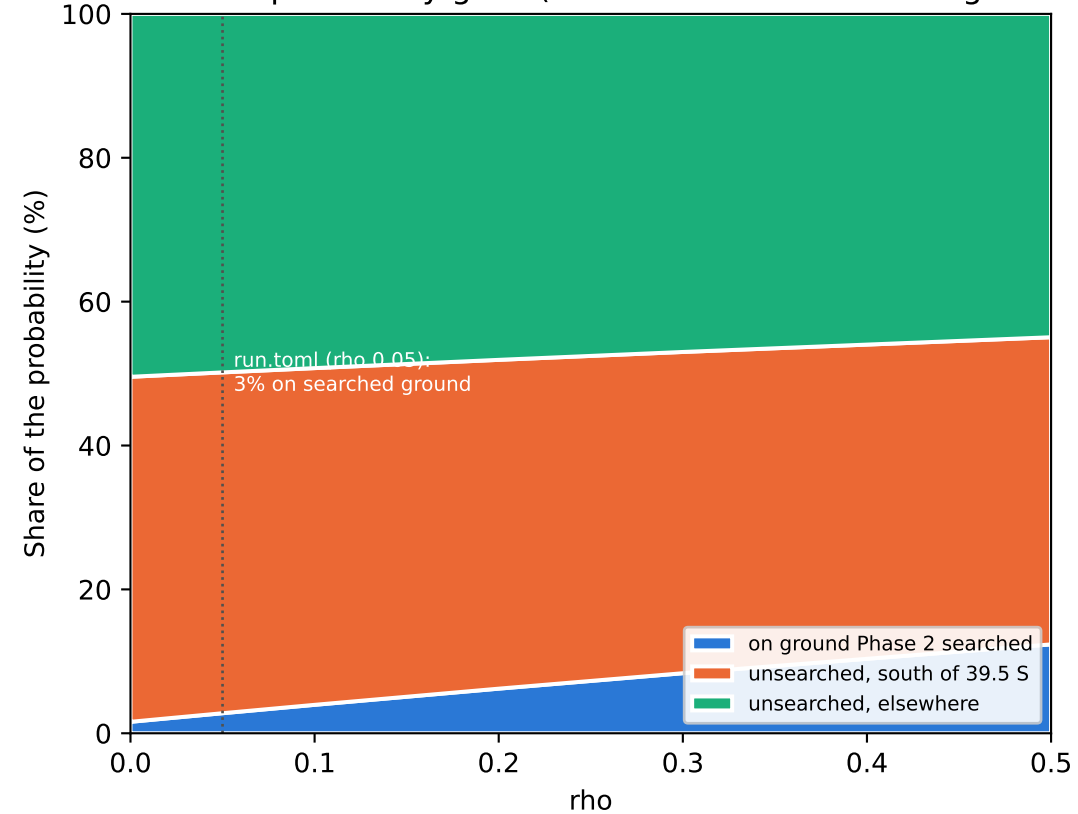


Impact latitude with the Phase 2 and Bluefin searches as evidence

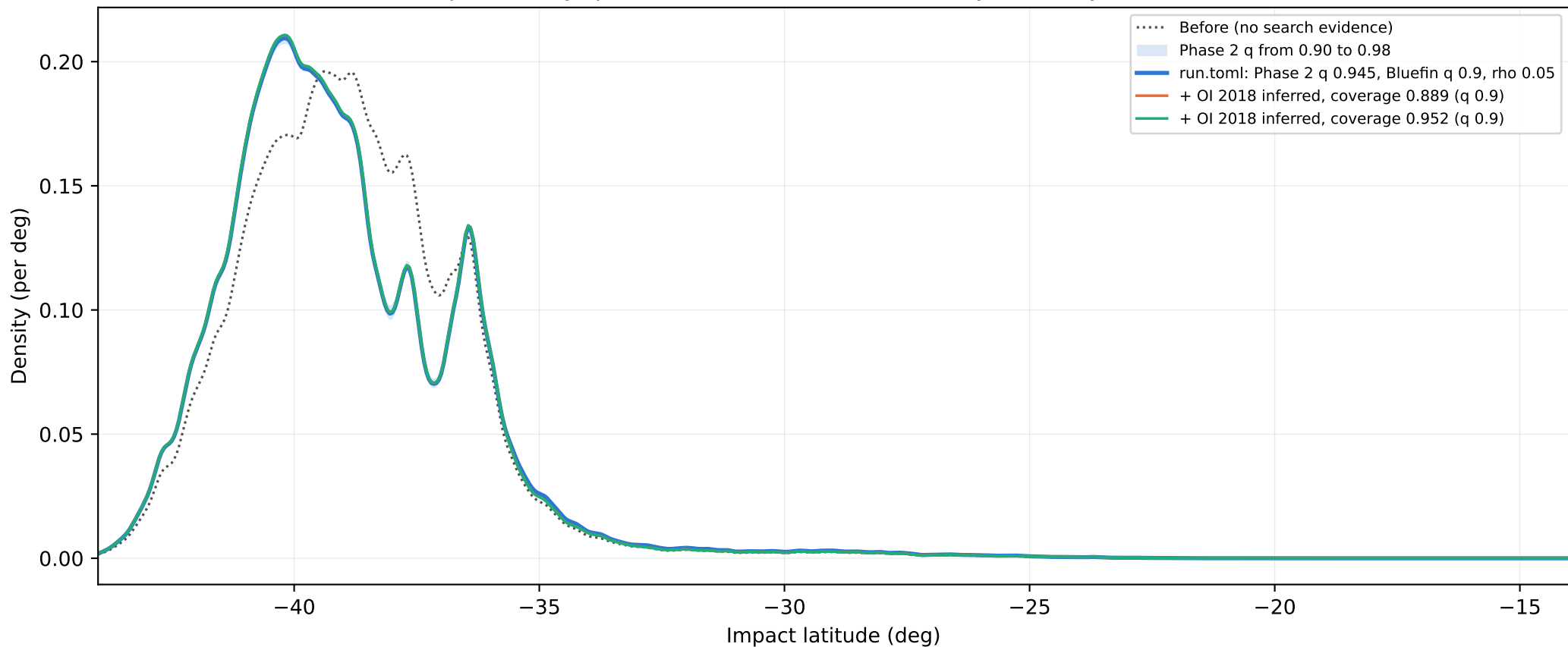


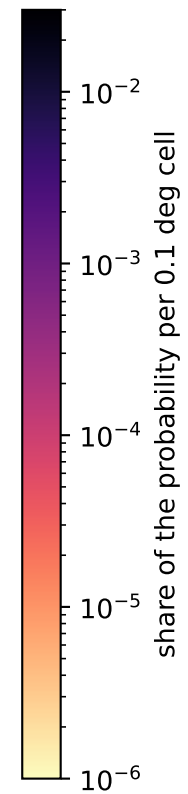
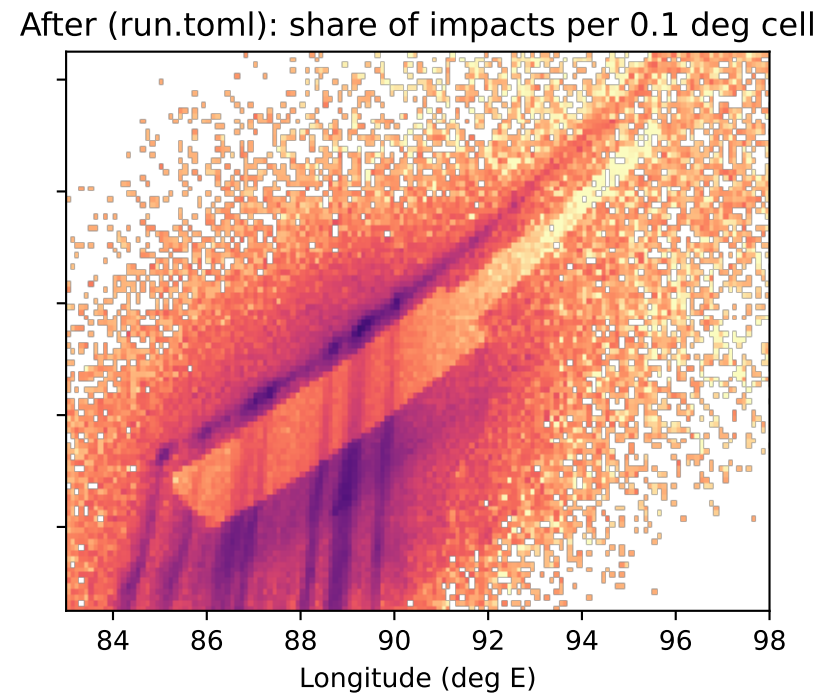
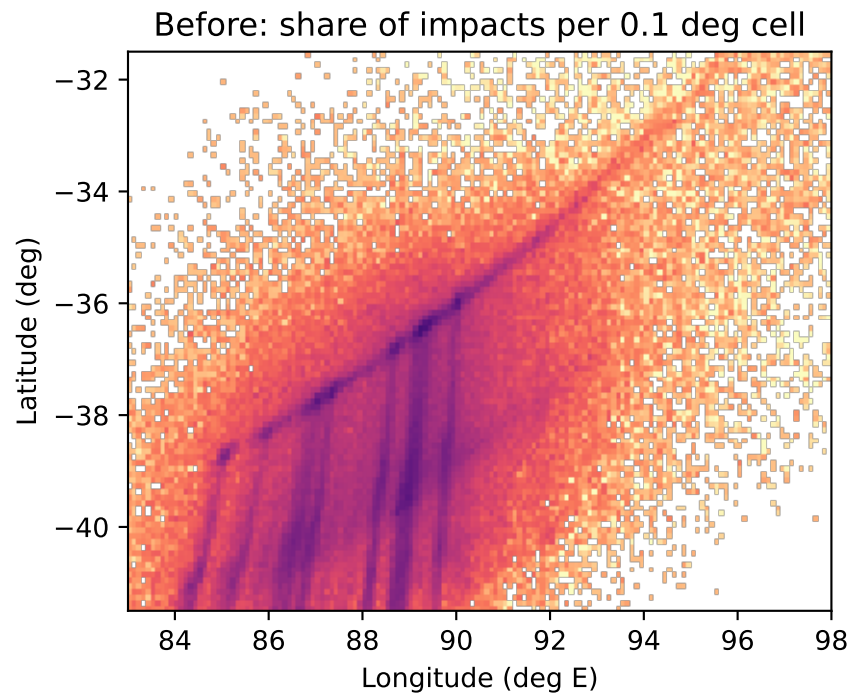
Where the probability goes (before: 21% on searched ground)



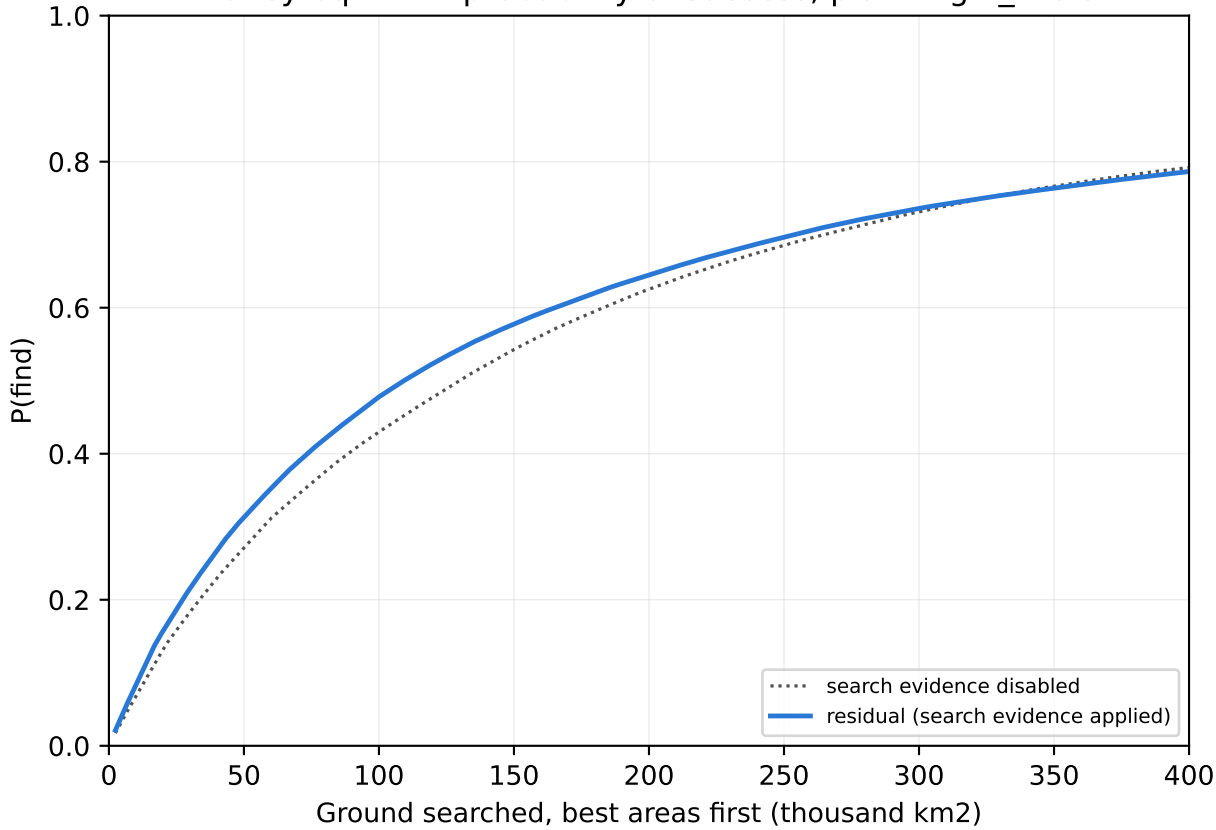
rho is the chance that the wreck could not have been found even where the sonar looked: hidden by terrain, buried, or imaged but dismissed. ATSB's detection ratings (q) cover data quality; rho is what they do not. At rho 0, 2% stays on searched ground; 79.0% of the impacts before the search lie off it.

Detection probability q , and the inferred Ocean Infinity 2018 layer (labelled variant)

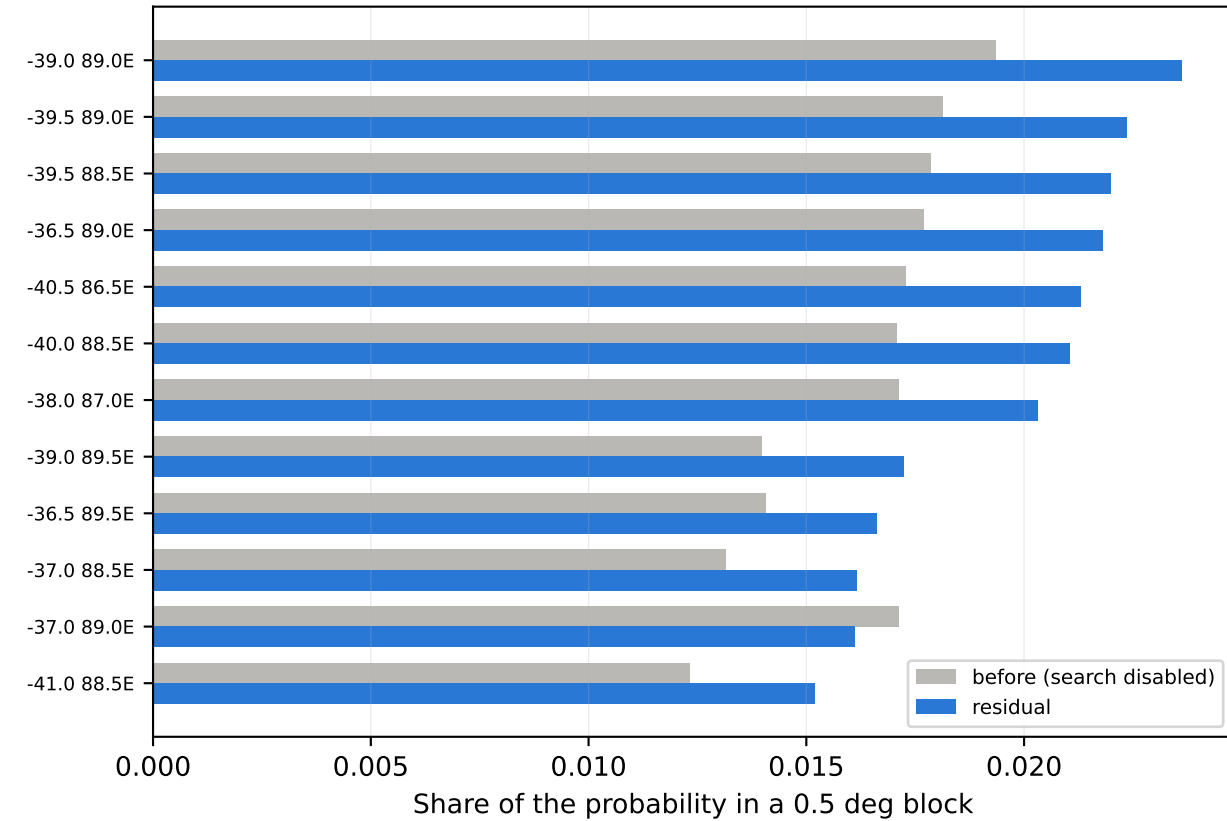




Davey eq. 11.2: probability of success, planning P_D 0.9



The best candidate areas, and what the search did to them



Seabed search evidence: numbers

IMPACTS FROM END-OF-FLIGHT MODULE END-OF-FLIGHT; 00:19 DATA OPTION NONE

Run eof-smoke8-4-29566: 8 replicates, 274,416 impacts; before: median -38.97, 95% -42.48 to -34.06; split-half 0.937; Kish ESS 255,140; on Phase 2 coverage 0.210

Mass removed at rho 0 (1 - evidence):

alone: bluefin-2014	0.0000
adding, in order: bluefin-2014	0.0000
alone: phase2-2014-2017	0.1983
adding, in order: phase2-2014-2017	0.1983
alone: ocean-infinity-2018	0.0049
adding, in order: ocean-infinity-2018	0.2032

scenario	Z	median	q025	q975	N of 33S	S of 39.5	on P2	ESS	split	vs before
run.toml	0.8116	-39.37	-42.60	-33.49	0.022	0.475	0.028	211,004	0.903	0.905
rho 0	0.8017	-39.40	-42.61	-33.45	0.022	0.481	0.016	206,302	0.901	0.899
rho 0.02	0.8057	-39.39	-42.61	-33.47	0.022	0.479	0.021	208,206	0.902	0.902
rho 0.1	0.8215	-39.35	-42.60	-33.53	0.022	0.470	0.040	215,504	0.905	0.911
rho 0.2	0.8414	-39.29	-42.58	-33.60	0.021	0.460	0.062	223,849	0.910	0.923
rho 0.3	0.8612	-39.25	-42.57	-33.67	0.021	0.451	0.083	231,248	0.914	0.934
rho 0.5	0.9008	-39.15	-42.54	-33.79	0.020	0.434	0.124	243,012	0.922	0.955
Phase 2 q 0.90	0.8206	-39.35	-42.60	-33.53	0.022	0.471	0.038	215,085	0.905	0.911
Phase 2 q 0.98	0.8046	-39.39	-42.61	-33.46	0.022	0.479	0.019	207,716	0.901	0.901
Phase 2 split, shared misses	0.8103	-39.38	-42.61	-33.49	0.022	0.476	0.026	210,375	0.903	0.904
Phase 2 split, independent misses	0.8091	-39.38	-42.61	-33.48	0.022	0.477	0.025	209,782	0.902	0.904
+ OI 2025-26 inferred, outboard band only	0.8110	-39.37	-42.61	-33.49	0.022	0.476	0.028	210,882	0.903	0.906
+ OI 2018 inferred, coverage 0.889	0.8070	-39.39	-42.61	-33.83	0.019	0.478	0.028	209,833	0.903	0.907
+ OI 2018 inferred, coverage 0.952	0.8067	-39.39	-42.61	-33.85	0.019	0.478	0.028	209,695	0.903	0.908

Residual PDF view. Source posterior: eof-smoke8-4-29566, 00:19 option none; search evidence

applied with run.toml. Davey eq. 11.2 at a planning P_D of 0.9 on 0.5 deg blocks:

rank	candidate area	residual mass	P(find)	km2	same block, search disabled
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1	-39.00	89.00	E	0.50	deg	0.0236	0.0213	2411	0.0194
2	-39.50	89.00	E	0.50	deg	0.0224	0.0201	2394	0.0181
3	-39.50	88.50	E	0.50	deg	0.0220	0.0198	2394	0.0178
4	-36.50	89.00	E	0.50	deg	0.0218	0.0196	2493	0.0177
5	-40.50	86.50	E	0.50	deg	0.0213	0.0192	2359	0.0173
6	-40.00	88.50	E	0.50	deg	0.0210	0.0189	2377	0.0171
7	-38.00	87.00	E	0.50	deg	0.0203	0.0183	2444	0.0171
8	-39.00	89.50	E	0.50	deg	0.0172	0.0155	2411	0.0140

P(find) 25% needs the best 16 blocks, 38,523 km²

P(find) 50% needs the best 46 blocks, 109,554 km²

P(find) 75% needs the best 136 blocks, 324,248 km²